DS-2002: Data Systems

A Primer in Relational Database Design

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SQL Database Design

Understanding the Principles that Govern Database Structure

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Fundamental Structures: Enforcing Data Integrity

Essential Design Concepts & Database Objects Required for Enforcing Data Integrity

Entity Integrity

- Enforced by the **Table**
  - Entities (nouns):
    - People, Places and Things
  - Concrete: Employees, Customers, Products
  - Conceptual: Sales, Scenarios, etc.

Domain Integrity

- Enforced by the **Column**
  - Data Type definition:
    - Int, Decimal, Float, Char, Nchar, Varchar, Nvarchar, DateTime
  - Constraints:
    - Primary Key, Check, Unique, & Default

Relational Integrity

- Enforced by the **Foreign Key Relationship**
  - One-to-Many: Foreign key relates to Primary key
  - Many-to-Many: Primary keys relate to Foreign keys via a *Juncture table*

Database Normalization: The Normal Forms

There are other Normal Forms, but Resolving to 3rd NF is Considered Appropriate

First Normal Form (1NF)

- A tables columns must contain only atomic values; they may not contain multiple values
- **Ex:** a column named **telephone_number** may contain only one phone number.

Second Normal Form (2NF)

- The table must first satisfy the first normal form.
- The table must be free of partial dependencies; i.e., all columns that are not the Primary Key must depend on the Primary Key

Third Normal Form (3NF)

- The table must first satisfy both the first and second normal forms
- The table must be free of transitive dependencies; i.e., no column may depend on any column that is not the Primary Key.

Workload Characteristics: Form Follows Function

Two Essentially Incompatible Workloads… They Have a Contentious Relationship

Online Transaction Processing (OLTP)

- Characterized by a large volume of transactions each of which affect a small number of rows
- Online Sales, Bank Deposits & Transfers
- Highly Normalized Database Schema

Online Analytical Processing (OLAP)

- Characterized by a small volume of read transactions each of which affect a large number of rows
- Periodic Post-hoc Analysis *(What Happened?)*
- De-Normalized Multi-Dimensional Schema

! These two **don't** play well together: They contend for the same hardware resources!

Database Paradigms: Design Approaches

The Design Approach Accommodates the Workload Characteristic

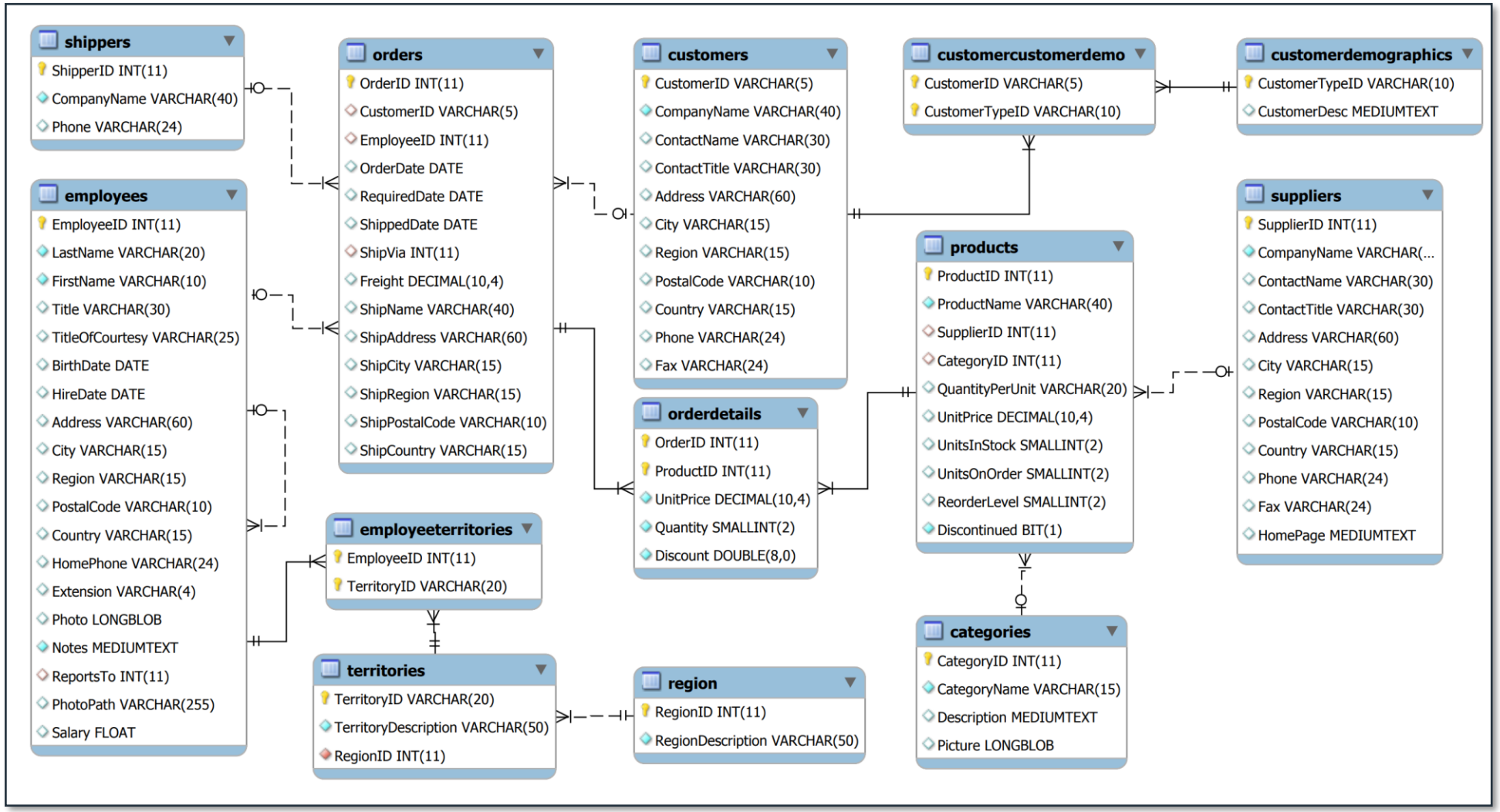
Normalized Relational Database:

- Optimized for Online Transaction Processing (OLTP) workloads
- Aims to Eliminate Data Redundancy and Minimize Storage Requirements
- **Complex:** Sacrifices User-Friendliness in Favor of Transactional Performance

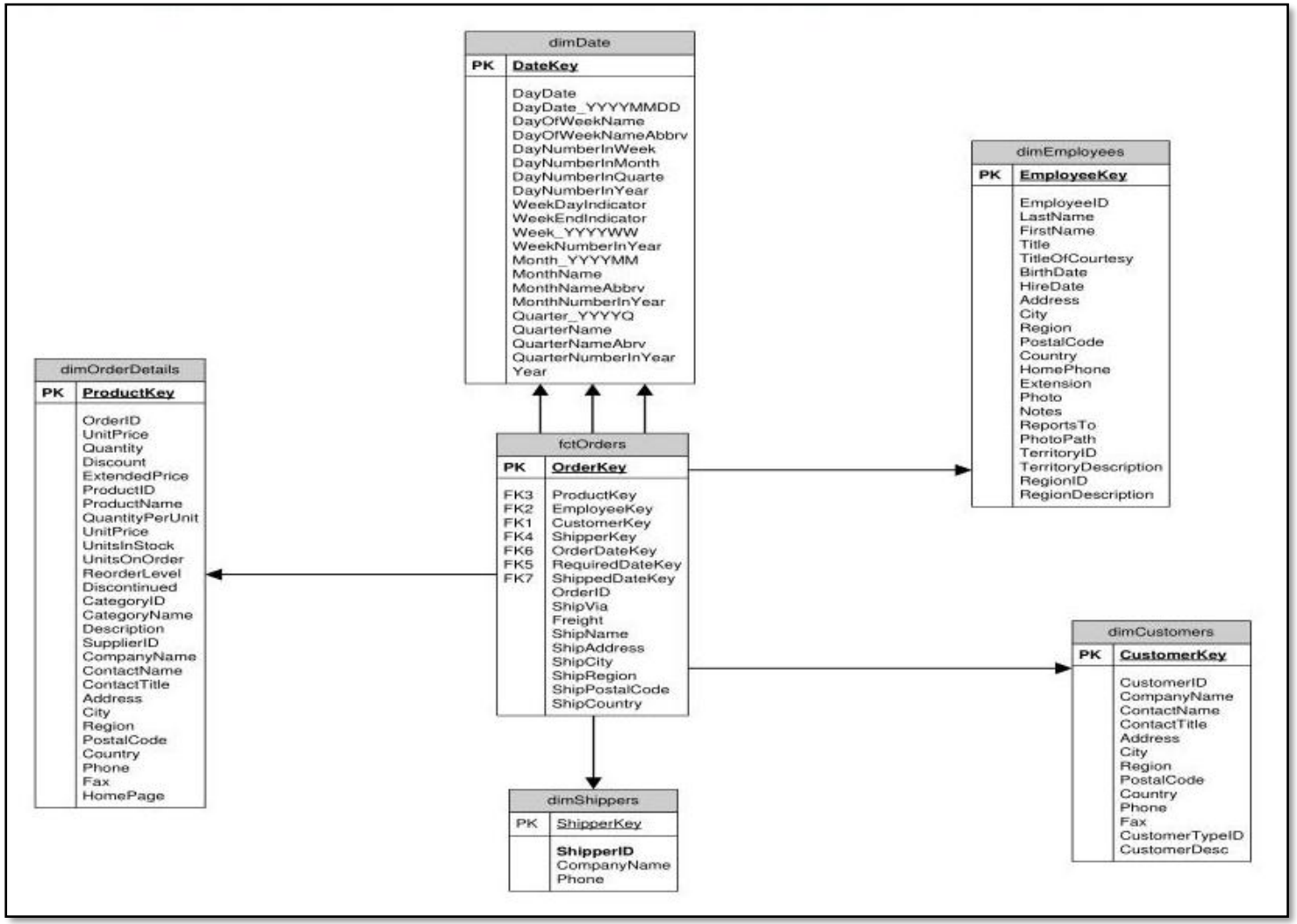
Multi-Dimensional Relational Database:

- Optimized for Online Analytical Processing (OLAP) workloads
- Aims to Optimize Query Performance and Provide an Intuitive User Experience
- **Simple:** Accepts Data Repetition in Favor of User-Friendliness and Improved Query Performance

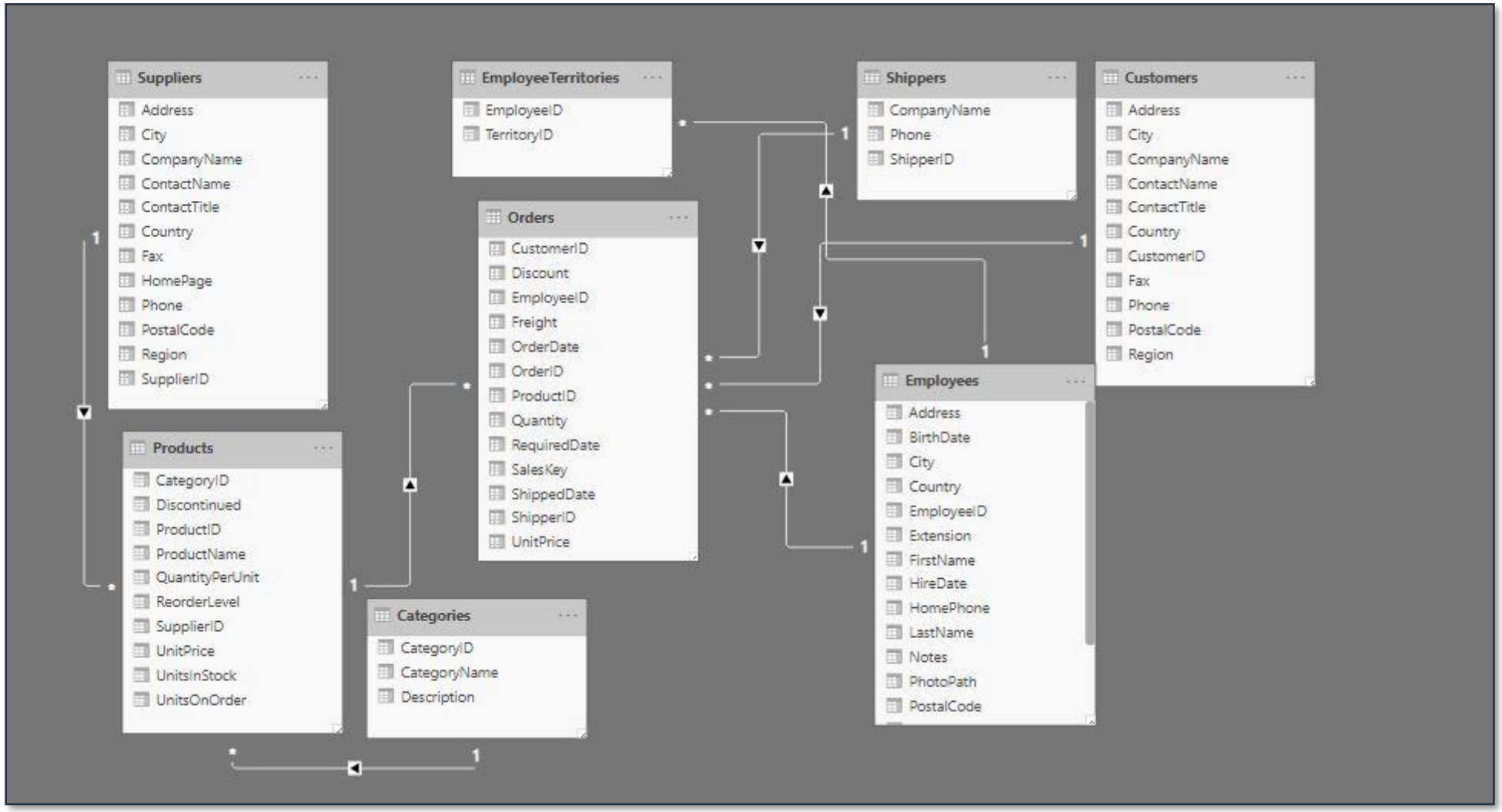
OLTP Database: Normalized Schema



OLAP Database: Multi-Dimensional (Star) Schema



OLAP Database: Snowflake Schema



Database Management System (DBMS) Internal Structures

Understanding the System Constructs that Optimize Data Storage and Retrieval

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Database Transactions: Enforcing Data Integrity

The **ACID** Properties of a Transaction Must be Enforced to Ensure the Integrity of a Database

Atomicity

- All operations in a transaction must be treated as a single [atomic] unit, and must either be fully committed (i.e., succeed) or be rolled-back (i.e., have no effect whatsoever).

Consistency

- Each transaction must guarantee the valid state of the database; both before and after the transaction is either committed or rolled-back.
- All defined rules, constraints, and relationships must be enforced and maintained.

Isolation

- Concurrent transactions must be completely independent; having no effect on each other.
- Transactions must remain invisible to each other until they are either committed or rolled-back.

Durability

- Once committed, the effects of each transaction must be permanently saved (written to disk); even if the system fails amidst their execution.
- A transaction log (aka, Journal file) serves as a write-ahead mechanism to ensure each committed transaction can be replayed to successfully recover the database to a consistent state in the event of a system failure.

Indexes: Quickly Finding a Needle in a Haystack

Physical Data Constructs that Optimize Locating Data on Storage Subsystems

Clustered Indexes

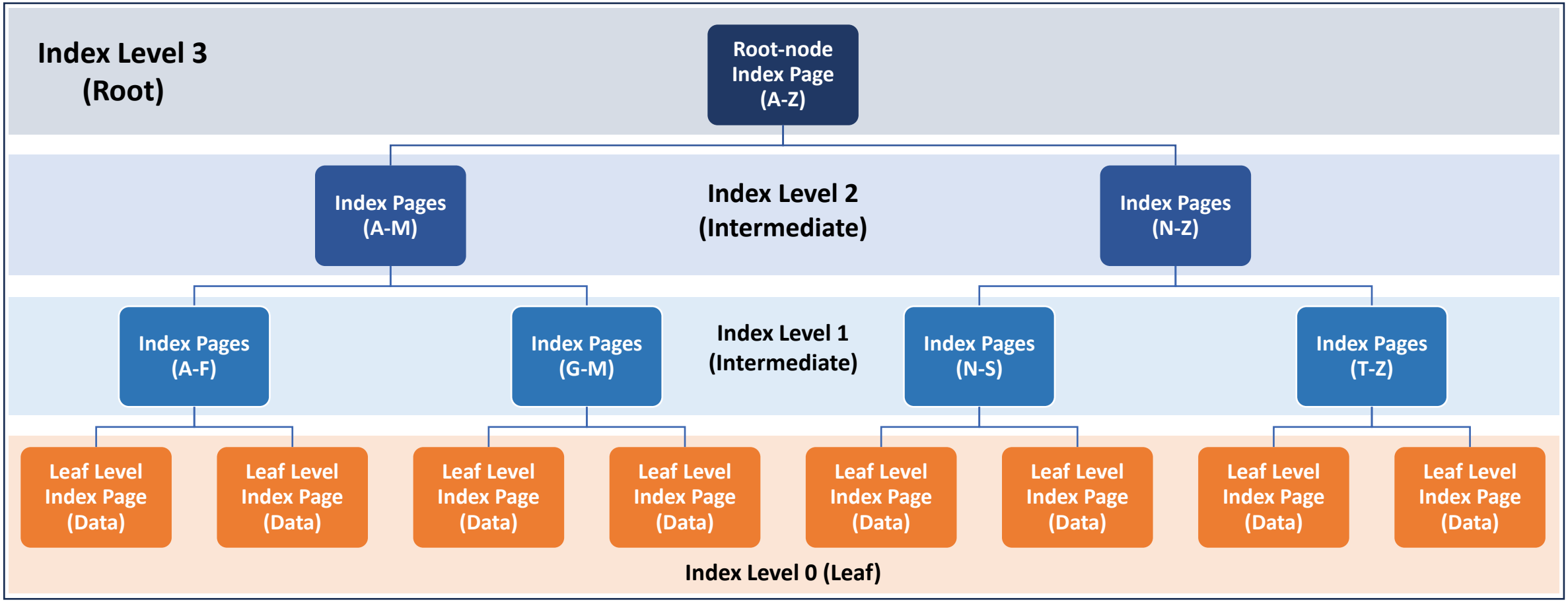
- BTree (Balanced Tree) Structure that determines the physical order of data in a table according to the Primary Key.
- Row-wise Structure; i.e., identifies the row(s) in which search terms exists.
- Analogous to a Dictionary, where each word is stored in alphabetical order with words categorized into sections according to their first letter (e.g., A, B).

Non-Clustered Indexes

- References the Clustered Index to optimize data retrieval without altering the physical order of the data.
- Row-wise Structure; i.e., identifies the row(s) in which search terms exists.
- Analogous to an Index located near the end of a textbook, where references to pages containing specific keywords are identified; organized alphabetically.

BTree (Balanced Tree): Physical Storage Structure

Organizes Data to Enable Applying Binary Logic for Quickly Locating Data on Disk



** Conceptual Representation of A BTree Index*

Q & A

An Overview of SQL Databases